Balanced Class-II coefficient jets, continuum convergence, and A7 reconstruction

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claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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Purpose and scope

This note closes the selected child

`A13-CLASSII-BALANCED-COEFFICIENT-JET-CONTINUUM-AND-A7-RECONSTRUCTION.`

It does so by replacing the impossible literal products XQ and XXQ by explicit high-against-root-shell homogeneous-chaos jets. The third- and fourth-chaos jets converge at the advertised Sobolev regularities and in every finite probability moment. A second-order dyadic reconstruction then recovers the full rational production coefficient in the exact A7 covariance-normal scheme. The reconstruction retains the finite $\Sigma_\Lambda Q_\Lambda$ Wick-conversion term and all cross-contraction corrections.

The result is deliberately scoped. The torus has $L = 16$; the Gaussian is the three-complex-component, six-real A1 production field with covariance symbol bounded by $\langle k\rangle^{-4}$; the density floor is fixed at 10^{-12}; and all components use one common real-even scalar regulator. The theorem is finite volume. It does not control an adapted random Cameron–Martin shift, prove the signed one-use estimate, prove a Nelson moment, construct an interacting measure, remove the floor or regulator, or change A13 from T4.

The construction follows the paracontrolled and reconstruction principles of Gubinelli–Imkeller–Perkowski (https://arxiv.org/abs/1210.2684) and Hairer (https://arxiv.org/abs/1303.5113), but all power counts and the finite-cutoff identification needed here are stated below.

1. Gaussian and dyadic conventions

Write $X = (X^1, \dots, X^6)$ on $\mathbb{T}^3_L$. The production covariance satisfies, entrywise and in operator norm,

$$|\widehat{C}^{ab}(k)| \le C\langle k\rangle^{-4}, \qquad \widehat{C}^{ab}(-k) = \widehat{C}^{ab}(k) \in \mathbb{R}. \tag{1.1}$$

For a common regulator r_Λ put $X_\Lambda = r_\Lambda(D)X$ and

$$Q^{ab}_{\Lambda,i} = \partial_i X^a_\Lambda \partial_i X^b_\Lambda - \Gamma^{ab}_{\Lambda,i}, \qquad \Gamma^{ab}_{\Lambda,i} = \mathbb{E}[\partial_i X^a_\Lambda \partial_i X^b_\Lambda]. \tag{1.2}$$

The subtraction in (1.2) is the exact covariance of the same regulator.

Let

$$S_j = \mathbf{1}_{\{\|k\|_\infty \le 2^j\}}(D), \qquad \Delta_j = S_j - S_{j-1}, \tag{1.3}$$

with one declared low block, and put $R_j = I - S_{j-2}$. Empty negative indices are absorbed into the low block. The root-shell tensor is $Q_{\Lambda,i,j} = \Delta_j Q_{\Lambda,i}$. Products below are exact Fourier convolutions. No Galerkin projection is inserted at an intermediate node; one common projection may be applied after the complete product.

The admitted regulator class consists of common scalar multipliers with

$$r_\Lambda(-k) = r_\Lambda(k) \in \mathbb{R}, \quad \sup_{\Lambda,k} |r_\Lambda(k)| < \infty, \quad r_\Lambda(k) \longrightarrow 1, \tag{1.4}$$

and either support $\|k\| \le C\Lambda$ or uniform Schwartz tails at scale Λ. These hypotheses give a common summable Fourier majorant. They include the pinned sharp cubes and the common-even A7 regulator class, but exclude component-split and non-even multipliers.

2. Balanced homogeneous-chaos jets

Let P_m denote projection onto homogeneous Wiener chaos m of the underlying A1 Gaussian. For component indices a, b, c, d define

$$\mathbb{J}^{c;ab}_{1,\Lambda,i} = \sum_j \mathsf{P}_3\Big[(R_j X^c_\Lambda) Q^{ab}_{\Lambda,i,j}\Big], \tag{2.1}$$

$$\mathbb{J}^{cd;ab,L}_{2,\Lambda,i} = \sum_j \mathsf{P}_4\Big[((R_j X^c_\Lambda)(R_j X^d_\Lambda)) Q^{ab}_{\Lambda,i,j}\Big], \tag{2.2}$$

$$\mathbb{J}^{cd;ab,R}_{2,\Lambda,i} = \sum_j \mathsf{P}_4\Big[(R_j X^c_\Lambda)((R_j X^d_\Lambda) Q^{ab}_{\Lambda,i,j})\Big]. \tag{2.3}$$

The letters L, R record the two previously pinned sharp-cube parenthesisations. Since all inner products are exact and non-aliased, multiplication is associative before P_4 and hence

$$\mathbb{J}^L_{2,\Lambda} = \mathbb{J}^R_{2,\Lambda}. \tag{2.4}$$

This equality is after complete reconstruction. It does not assert equality of individual Bony channels, and it fails if a cutoff is reinserted between the two multiplications.

The high factor $R_j X$ is load-bearing. If it is replaced by X, the summable zero and low modes of X multiply the $|n|^{-1}$ spectrum of Q; the result has only H^{-1-} regularity. Equations (2.1)–(2.3) remove exactly that low–high piece while preserving an exact reconstruction in Section 5.

3. Fourier convolution estimate

Lemma 1 (balanced convolution)**.** *Let $d = 3$, $a(k) = \langle k\rangle^{-4}$ and $b(k) = \langle k\rangle^{-2}$. With χ_j the sharp-cube annulus from (1.3), for every $\epsilon > 0$ the complete balanced kernels satisfy*

$$V_3(n) \le C_\epsilon \langle n\rangle^{-2+\epsilon}, \tag{3.1}$$

$$V_4(n) \le C_\epsilon \langle n\rangle^{-3+\epsilon}. \tag{3.2}$$

Here V_3, V_4 are the Fourier variances of (2.1) and (2.2), with arbitrary fixed component indices. The same bounds hold for cross-cutoff differences with a factor tending to zero under the corresponding weighted sum.

Proof. The derivative covariance is bounded by b. The standard three-dimensional convolution split into $|p| \le |n|/2$, $|n-p| \le |n|/2$, and the remaining region gives

$$(b * b)(r) \le C\langle r\rangle^{-1}. \tag{3.3}$$

Fix the root shell $|r| \asymp 2^j$. In (2.1) the value momentum is restricted to $|k| \gtrsim 2^j$. If $2^j \ll |n|$, then $k = n - r$ has size $|n|$ and the shell mass of (3.3) is $O(2^{2j})$; summing up to $2^j \asymp |n|$ gives $O(|n|^{-2})$. If $2^j \gtrsim |n|$, both k and r are at least of shell size and the j-term is $O(2^{-2j})$. Its tail is again $O(\langle n\rangle^{-2})$.

For (2.2), both value momenta are restricted below by the root scale. In the low-root region, convolving the two a factors costs at most $|n|^{-4}2^{-j}$, while the Q shell has mass $O(2^{2j})$. Summation to $2^j \asymp |n|$ gives $O(|n|^{-3})$. At and above the output scale, direct rescaling of the two value sums and the root sum gives $O(2^{-3j})$.

The finite overlap of annuli gives the same estimates for diagonal shell pairs. Cauchy–Schwarz with weights $2^{\epsilon|j-j'|}$ absorbs the off-diagonal shell pairs at the displayed arbitrarily small loss. Wick isometry adds only a finite component/contraction factor. For two cutoffs, at least one covariance multiplier is replaced by a difference. Pointwise convergence in (1.4), the preceding summable majorant, and dominated convergence prove the cross-cutoff assertion. $\square$

Theorem 1 (continuum balanced jets)**.** *Let $1/3 < \alpha < 1/2$ and $\kappa > 0$. Then the nets (2.1)–(2.3) are Cauchy for every finite p in*

$$\mathbb{J}_{1,\Lambda} \longrightarrow \mathbb{J}_1 \qquad \text{in } L^p(\Omega; H^{\alpha-1-\kappa}), \tag{3.4}$$

$$\mathbb{J}_{2,\Lambda}^{L,R} \longrightarrow \mathbb{J}_2 \qquad \text{in } L^p(\Omega; H^{2\alpha-1-\kappa}). \tag{3.5}$$

The two second-jet limits agree and are common to the regulator class (1.4).

Proof. For a Fourier variance $\langle n\rangle^{-\beta+\epsilon}$ in three dimensions, the H^s second moment is finite for $s < (\beta-3)/2$. Equation (3.1) therefore permits every $s < -1/2$, and (3.2) every $s < 0$. Since $\alpha < 1/2$, the exponents in (3.4)–(3.5) are strictly below those ceilings. The cross-cutoff part of the lemma proves L^2 Cauchy convergence. Hilbert-valued homogeneous-chaos hypercontractivity gives

$$\|Z\|_{L^p(\Omega;H^s)} \le C_m(p-1)^{m/2}\|Z\|_{L^2(\Omega;H^s)} \tag{3.6}$$

for chaos order $m = 3, 4$, proving every finite p. Equation (2.4) and dominated convergence give the remaining statements. $\square$

4. Complete lower-chaos conversion

Put $Z_j^c = R_j X^c$ and $Q_j^{ab} = \Delta_j Q^{ab}$. At every finite cutoff, Wick's formula gives the exact decompositions

$$Z_j^c Q_j^{ab} = \mathsf{P}_3[Z_j^c Q_j^{ab}] + \mathsf{P}_1[Z_j^c Q_j^{ab}], \tag{4.1}$$

$$\begin{aligned} Z_j^c Z_j^d Q_j^{ab} &= \mathsf{P}_4[Z_j^c Z_j^d Q_j^{ab}] + \mathsf{P}_2^{\text{cross}}[Z_j^c Z_j^d Q_j^{ab}] \\ &\quad + \Sigma_j^{cd} Q_j^{ab} + \mathsf{P}_0^{\text{cross}}[Z_j^c Z_j^d Q_j^{ab}], \end{aligned} \tag{4.2}$$

where $\Sigma_j^{cd} = \mathbb{E}[Z_j^c Z_j^d]$. The first correction has two value–derivative contractions. The cross part in (4.2) has four second-chaos and two zero-chaos contractions. Together with $\Sigma_j Q_j$, raw ZZQ has five second-chaos and two zero-chaos families. For the recursive parenthesisation $Z\mathsf{P}_3[ZQ]$ there are three second-chaos and no zero-chaos contractions. These are the previously proved 2, $4+2$, $5+2$, and $3+0$ forests, now attached to the balanced root-shell multiplier.

The cross pieces in (4.1)–(4.2) need not vanish shell by shell. They are therefore retained, rather than replaced by the global same-point parity identity. In a scalar two-variable notation with $\mathbb{E}Z^2 = \sigma$, $\mathbb{E}Y^2 = \gamma$, and $\mathbb{E}ZY = k$,

$$Z(Y^2 - \gamma) =: ZYY : +2kY, \tag{4.3}$$

$$Z^2(Y^2 - \gamma) =: ZZYY : +\sigma(Y^2 - \gamma) + 4k : ZY : +2k^2. \tag{4.4}$$

Equation (4.4) displays the finite ΣQ term. Deleting it changes the finite-cutoff random variable and is not a change of notation.

5. Exact rational coefficient reconstruction

The six-real production matrix is

$$B(X) = \sum_A \{ a p_A p_A^T + b(p_A v_A^T + v_A p_A^T) + c v_A v_A^T \}, \tag{5.1}$$

where

$$p_A = 2S_A X, \quad v_A = 2(S_A - q_A I)X, \quad q_A = \frac{X^T S_A X}{X^T X + \epsilon_\rho}. \tag{5.2}$$

The fixed positive floor makes B smooth; its derivatives have polynomial growth with constants depending on the floor.

For a root shell j write $x_j = S_{j-2}X$ and $z_j = R_j X$. For any matrix coefficient F equal to B or

$$F_h(x) = B(x+h) - B(x) \tag{5.3}$$

with deterministic $h \in H^2$, Taylor's formula is the exact identity

$$F(x_j + z_j) = F(x_j) + DF(x_j)[z_j] + \frac{1}{2} D^2 F(x_j)[z_j, z_j] + \mathcal{R}_{3,j}^F. \tag{5.4}$$

Multiplying by Q_j, substituting (4.1)–(4.2), and summing j gives the finite-cutoff reconstruction chart

$$\begin{aligned} F(X) : Q = {} & \sum_j F(x_j) : Q_j \\ & + \sum_j DF(x_j) : \{ \mathsf{P}_3[z_j Q_j] + \mathsf{P}_1[z_j Q_j] \} \\ & + \frac{1}{2} \sum_j D^2 F(x_j) : \{ \mathsf{P}_4[z_j z_j Q_j] + \mathsf{P}_2^{\text{cross}}[z_j z_j Q_j] \\ & \qquad\qquad\qquad\qquad + \Sigma_j Q_j + \mathsf{P}_0^{\text{cross}}[z_j z_j Q_j] \} \\ & + \sum_j \mathcal{R}_{3,j}^F : Q_j. \end{aligned} \tag{5.5}$$

All tensor indices are contracted exactly as in (5.1). Formula (5.5) is an identity, not an asymptotic expansion. It uses neither a finite Hermite truncation of B nor a new physical counterterm.

For the quadratic test $F^{cd}(X) = X^c X^d$, the low chart contains $\Sigma_{<j}^{cd} Q_j$ and (4.2) contains $\Sigma_j^{cd} Q_j$. Their complete sum is precisely

$$\Sigma_\Lambda^{cd} Q_\Lambda^{ab}, \tag{5.6}$$

the finite conversion term from the preceding forest theorem. Thus (5.5) cannot silently Wick-order the coefficient itself.

Theorem 2 (A7-scheme reconstruction)**.** *Assume*

$$\frac{1}{3} < \alpha < \frac{1}{2}, \qquad 0 < \kappa < 3\alpha - 1. \tag{5.7}$$

The dyadic chart (5.5), interpreted as the standard modelled-distribution reconstruction from $Q, \mathbb{J}_1, \mathbb{J}_2$ and the displayed lower chaoses, is continuous in those model norms. Its regularised values converge in every finite probability moment in $H^{-1-\kappa}$. For $F = B$ the limit is exactly the A7 covariance-normal composite

$$B(X) : (\partial_i X \otimes \partial_i X - \Gamma_i). \tag{5.8}$$

For deterministic $h \in H^2$, $F = F_h$ gives the translated coefficient term in the same scheme. No intermediate projection or extra divergent local counterterm appears.

Proof. At scale j, subtract the order-two expansion (5.4). The remainder gains 3α, while Q_j has order $-1-\kappa$. Condition (5.7) makes the sewing exponent $3\alpha-1-\kappa$ positive. The order-one and order-two increments are represented by (2.1)–(2.3); their model bounds are (3.4)– (3.5). The lower-chaos terms in (4.1)–(4.2) are fixed Gaussian contractions of the same kernels and are grouped with the Taylor levels in (5.5). The usual dyadic telescoping estimate therefore makes the reconstruction map continuous and unique at regularity $-1-\kappa$.

Every regularised object is smooth. For it, the reconstruction axiom is ordinary multiplication, so (5.5) equals its left-hand side exactly. With $F = B$, that left-hand side is (5.8) with the same-regulator covariance (1.2), hence is exactly the A7 finite-cutoff prescription. A7 proves its common-regulator limit. Continuity of the present chart and Theorem 3.2 identify the two limits and show explicitly where all lower chaoses sit.

For (5.3), approximate deterministic $h \in H^2$ by smooth functions. The three-dimensional embedding $H^2 \hookrightarrow C^\beta$ for every $\beta < 1/2$, the fixed-floor derivative bounds on B, and the same chart estimate give a locally Lipschitz limit. Passing through the smooth approximants proves the translated statement. $\square$

At the production values $\alpha = 0.4$, $\kappa = 0.1$, the first and second targets are -0.7 and -0.3, below the variance ceilings -0.5 and 0. The reconstruction remainder has positive exponent $3\alpha-1-\kappa = 0.1$.

6. Executable adversarial audit

The primary program derives the exponent chain from $d = 3$ and the A1 q^{-4} covariance, checks exact sparse non-aliased associativity, evaluates the complete 2, $4+2$, $5+2$, and $3+0$ forest identities with a nonzero ΣQ deletion control, and differentiates the full six-real rational matrix analytically. The independent program does not import it: it uses zero-padded dense Fourier convolution, Gauss–Hermite projection, and complex-step versus five-point differentiation for the production matrix. The integrated verifier pins every authority and source hash, reruns both children, and checks the claim/gate boundary.

The finite fixtures test algebra, signs, conventions, and hardcode masking. They do not infer continuum convergence from a fitted slope. The continuum claim rests on Lemma 3.1, Theorem 3.2, and the reconstruction proof.

Devil's-advocate

1. **DISMISSED: the literal full product already has the target regularity.** Its low value modes preserve the $|n|^{-1}$ variance of Q. The high-against-root restriction in (2.1)–(2.3) is essential.

2. **VALID WITH MITIGATION: homogeneous-chaos projection loses the original rational coefficient.** Projection is used only for the two model jets. Formula (5.5) is an exact second-order reconstruction and retains the Taylor remainder, every lower chaos, and the full nonpolynomial B.

3. **VALID WITH MITIGATION: parity makes all lower chaoses zero.** Global same-point value–derivative covariance is zero, but a root-shell selector need not preserve that cancellation. The P_1, $\mathsf{P}_2^{\mathrm{cross}}$, and $\mathsf{P}_0^{\mathrm{cross}}$ terms are displayed and retained.

4. **DISMISSED: finite ΣQ may be deleted because it is not divergent.** Equation (4.4) and the quadratic test (5.6) show that deletion changes the finite-cutoff composite. Finiteness is not zero.

5. **UPHELD AGAINST INTERMEDIATE PROJECTION: both trees agree after projecting every inner product.** They need not. Equation (2.4) assumes exact convolution and at most one common root projection.

6. **VALID WITH MITIGATION: the theorem is regulator universal.** It is universal only in the declared common real-even scalar class with a uniform Fourier majorant and exact same-regulator covariance subtraction.
7. **UPHELD AGAINST PROMOTION: this closes the signed one-use or Nelson theorem.** It supplies the enhanced coefficient model and exact A7 bridge only. Adapted-shift stochastic coercivity is the next gate; A13 stays T4.

Result footer

Result ID:	A13-CLASSII-BALANCED-COEFFICIENT-JET-CONTINUUM-AND-A7-RECONSTRUCTION
Claim:	A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Precise statement:	Balanced P3/P4 root-shell jets converge in every finite probability moment at H^(alpha-1-kappa) and H^(2alpha-1-kappa). A second-order dyadic chart reconstructs the full rational translated coefficient in the exact A7 covariance-normal scheme, retaining Sigma Q and every lower-chaos correction.
Parameter range:	1/3 < alpha < 1/2; 0 < kappa < 3 alpha - 1 for the reconstruction; alpha=0.4, kappa=0.1 is the pinned audit point.
Regulators:	Common bounded real-even scalar multipliers, pointwise to one, compact support or uniform Schwartz tail; same-regulator Gamma; exact/dealiased products.
Dependencies:	A1 production functional; A7 covariance-normal composite; A13 universal Q and finite forest results.
Tier before / after:	T4 / T4 (no tier change).
Falsifiers:	Wrong -2/-3 variance exponents; failed cross-cutoff Cauchy estimate; missing lower chaos or Sigma Q; intermediate-projection associativity; rational chart mismatch; authority/hash/assertion failure.
No-overclaim:	No adapted random-shift form bound, one-use estimate, Nelson moment, Gibbs measure, floor/regulator removal, infinite volume, phase transition, BCC, T5, T6, T7.
Next required action:	Prove the strict-past signed Q/current/positive-square/Cartan-curvature one-use estimate using this model.